



Future Interns

Cyber Security Task 2 (2026)

Phishing Email Detection and Awareness Report

Security Awareness Analysis for Corporate Environments

Prepared by Ebasa Getu

Report Metadata

Report Type	Phishing Email Detection and Awareness
Date	June 2026
Methodology	Email Header Analysis, Domain Reputation Check, URL Inspection, Content Analysis
Analyst	Ebasa Getu

Executive Summary

This report presents a detailed analysis of three phishing email samples commonly encountered in corporate environments. Each email was analyzed for header anomalies, sender domain reputation, linguistic indicators of phishing, and technical red flags. The goal is to educate employees on recognizing phishing attacks and to provide clear, actionable prevention guidelines.

Of the three emails analyzed, two were classified as Phishing and one as Suspicious. All exhibited multiple red flags including spoofed sender domains, urgency-based language, generic greetings, and suspicious links.

Email Risk Summary

#	Subject	Sender	Risk	Attack Type
1	Urgent: Your Account Will Be Locked	"Security Team" <security@secure-account-verify.com>	Phishing	Credential Harvesting
2	Invitation: Annual Team Building Event	"HR Department" <hr@conference-invite.co>	Phishing	Targeted HR Impersonation
3	Action Required: Update Your Microsoft 365 Credentials	"IT Support" <support@microsoft-verify.net>	Suspicious	Brand Impersonation

Sample 1: Urgent: Your Account Will Be Locked

From	"Security Team" <security@secure-account-verify.com>
Risk Classification	Phishing

Email Headers

```
Delivered-To: employee@company.com
Received: from mail.secure-account-verify.com (172.67.215.175)
  by mx.company.com with ESMTTP
Return-Path: <bounce@secure-account-verify.com>
From: "Security Team" <security@secure-account-verify.com>
Reply-To: "Security Team" <verify@secure-account-verify.com>
Subject: Urgent: Your Account Will Be Locked
DKIM-Signature: v=1; a=rsa-sha256; d=secure-account-verify.com;
Date: Mon, 15 Jun 2026 09:24:17 +0000
```

Email Body

```
Dear User,

We noticed suspicious activity on your account.
To avoid account suspension, please verify your details immediately.

Verify Now: http://secure-account-verify.com/login

Failure to verify within 24 hours will result in permanent account lock.

Regards,
Security Team
```

Identified Phishing Indicators

1. Fake sender domain - secure-account-verify.com is not the company real domain
2. Generic greeting - Dear User instead of using recipient real name
3. Urgency and fear-based language - Your Account Will Be Locked, 24 hours
4. Suspicious link - http (not HTTPS), domain unrelated to the company
5. Reply-To address differs from From address - common email spoofing
6. Return-Path points to suspicious domain, not the company mail server
7. DKIM domain does not match the legitimate organization

Analysis

This is a classic credential harvesting phishing attack. The attacker impersonates the company security team and creates a false sense of urgency to trick recipients into clicking the malicious link. The domain secure-account-verify.com is not owned by the company - it was registered recently and is hosted behind Cloudflare to evade detection. The link leads to a fake login page designed to steal usernames and passwords.

Sample 2: Invitation: Annual Team Building Event

From	"HR Department" <hr@conference-invite.co>
Risk Classification	Phishing

Email Headers

Delivered-To: employee@company.com
Received: from mail.conference-invite.co (185.234.72.18)
by mx.company.com with ESMT
Return-Path: <bounce@conference-invite.co>
From: "HR Department" <hr@conference-invite.co>
Reply-To: "HR Department" <rsvp@conference-invite.co>
Subject: Invitation: Annual Team Building Event
Date: Mon, 15 Jun 2026 14:30:22 +0000

Email Body

Dear Employee,

You have been selected to attend the Annual Team Building Event!
We have reserved a spot for you at the luxury resort.
Please confirm your attendance and provide your details below:

Confirm here: <http://conference-invite.co/confirm>

Please respond within 48 hours to secure your spot.

Best,
HR Department

Identified Phishing Indicators

1. Suspicious sender domain - conference-invite.co is not the company HR domain
2. Link points to a .co domain (not .com) - commonly used by phishers
3. Requests personal details through an external link rather than internal portal
4. Generic closing with no specific contact person
5. No phone number or alternative contact provided
6. Domain registered recently - typical of phishing campaigns

Analysis

A targeted phishing email disguised as an HR invitation. The attacker leverages the excitement of a team-building event to lower the recipient guard. The .co domain mimics a legitimate corporate domain. The external form would capture the employee login credentials and personal information.

Sample 3: Action Required: Update Your Microsoft 365 Credentials

From	"IT Support" <support@microsoft-verify.net>
Risk Classification	Suspicious

Email Headers

Delivered-To: employee@company.com
Received: from mail.microsoft-verify.net (45.33.32.156)
by mx.company.com with ESMTP
Return-Path: <bounce@microsoft-verify.net>
From: "IT Support" <support@microsoft-verify.net>
Reply-To: <support@microsoft-verify.net>
Subject: Action Required: Update Your Microsoft 365 Credentials
Date: Mon, 15 Jun 2026 11:05:43 +0000

Email Body

Dear Employee,

Our system indicates that your Microsoft 365 account requires immediate verification.

Failure to update your credentials may result in loss of access to email, Teams, and company files.

Click below to verify your account:
<https://microsoft-verify.net/office365/login>

Thank you,
IT Support Team

Identified Phishing Indicators

1. Domain microsoft-verify.net mimics Microsoft but is not official
2. Threat of service disruption - loss of access creates urgency
3. Generic greeting - Dear Employee
4. Link leads to a lookalike domain (typosquatting)
5. No official Microsoft branding in the email body
6. No ticket number or reference ID for the supposed issue

Analysis

A brand impersonation phishing attack targeting Microsoft 365 credentials. The domain microsoft-verify.net uses the legitimate Microsoft name to build trust. The email creates anxiety about losing access to critical business tools. Classified as Suspicious because the domain might be a compromised legitimate site rather than a freshly registered phishing domain.

Prevention and Awareness Guidelines

How Phishing Attacks Work

Phishing attacks rely on psychological manipulation rather than technical exploits. Attackers create emails that appear legitimate and trigger emotional responses - urgency, fear, curiosity, or excitement - to bypass rational decision-making. The goal is to trick the recipient into clicking a malicious link, downloading an infected attachment, or sharing sensitive information.

Do's for Employees

1. Verify sender addresses carefully. Check the full email address, not just the display name.
2. Hover over links before clicking. Move your mouse over a link to see the actual URL.
3. Check for HTTPS. Legitimate login pages use HTTPS (the padlock icon).
4. Use multi-factor authentication (MFA). Even if your password is stolen, MFA can prevent unauthorized access.
5. Report suspicious emails to your IT security team immediately.
6. Verify urgent requests through another channel. Call the department directly to confirm.
7. Keep software updated. Regular updates patch security vulnerabilities.

Don'ts for Employees

1. Do not click links in unsolicited emails. Navigate to websites manually.
2. Do not enter credentials on pages reached via email links.
3. Do not download attachments from unknown senders.
4. Do not ignore warning signs. Generic greetings, spelling errors, and unusual urgency are red flags.
5. Do not share OTPs, passwords, or personal information via email.
6. Do not forward suspicious emails to others. Report it to IT instead.
7. Do not disable security warnings in your browser or email client.

Phishing Indicator Checklist

Indicator	What to Look For
Sender Address	Does the email domain match the organization?
Greeting	Is it generic (Dear User) or personalized?
Urgency	Does it pressure you to act immediately?
Links	Do the hover URLs match the claimed destination?
Attachments	Were you expecting them? Is the file type suspicious?
Spelling and Grammar	Are there typos or awkward phrasing?
Reply-To Address	Does it differ from the From address?
Request for Credentials	Does it ask you to log in or verify?
Domain Reputation	Is the domain newly registered?

Tools Used

Tool	URL / Command
Email Header Analyzer	toolbox.googleapps.com/apps/messageheader/
MXToolbox	mxtoolbox.com/EmailHeaders.aspx
Domain WHOIS	whois.domaintools.com

Tool	URL / Command
DNS Lookup	host / dig commands
Link Analysis	Browser DevTools, curl

Methodology

1. Header Inspection: Extract and analyze email headers to verify sender authenticity and SPF/DKIM/DMARC records.
2. Domain Reputation Check: Verify whether the sender domain is legitimate, newly registered, or associated with known phishing.
3. URL Analysis: Inspect embedded links for discrepancies between the displayed text and actual URL.
4. Content Analysis: Evaluate the email body for linguistic indicators: urgency, fear tactics, generic greetings.
5. Risk Classification: Assign a risk level (Safe / Suspicious / Phishing) based on the number and severity of indicators.